

Wireless Networking

1. List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

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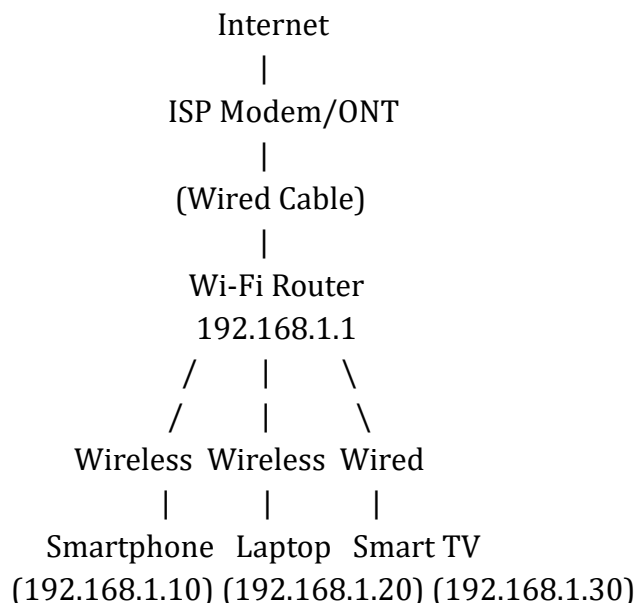
- Popular Wi-Fi Standards

Wi-Fi Standard	Frequency Band	Maximum Speed	Key Difference
802.11n (Wi-Fi 4)	2.4 GHz and 5 GHz	Up to 600 Mbps	First standard to support both 2.4 GHz and 5 GHz bands.
802.11ac (Wi-Fi 5)	5 GHz only	Up to 6.9 Gbps	Faster speeds and less interference due to 5 GHz operation.
802.11ax (Wi-Fi 6)	2.4 GHz and 5 GHz (and 6 GHz in Wi-Fi 6E)	Up to 9.6 Gbps	Better performance in crowded networks and higher efficiency.

2. Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.

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- Network Diagram



- **Labels**

Wired Connection:

- Internet → ISP Modem → Wi-Fi Router

Wireless Connection:

- Router → Smartphone
- Router → Laptop

Wired Connection:

- Router → Smart TV (Ethernet Cable)

3. Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.

ANS : Available Wi-Fi Networks

SSID	Signal Strength	Security Type
Home_WiFi	-42 dBm	WPA3
JioFiber_5G	-58 dBm	WPA2/WPA3
Neighbor_WiFi	-75 dBm	WPA2

- **Which Network Would I Choose?**

- I would choose Home_WiFi because:

1. It has the strongest signal (-42 dBm).
2. It uses WPA3 security, which is more secure than WPA2.
3. A stronger signal provides better speed and stability.

- **Signal Strength Guide**

Signal Strength	Quality
-30 to -50 dBm	Excellent
-51 to -67 dBm	Good
-68 to -80 dBm	Fair
Below -80 dBm	Weak

4. Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

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- **What is SSID Hiding?**

- SSID (Service Set Identifier) is the name of a Wi-Fi network.
- When SSID hiding is enabled, the router does not broadcast the network name publicly.
- Users must manually enter the Wi-Fi name to connect.

- **Normal Example**

- **Available Networks:**

1. Home_WiFi
2. JioFiber_5G
3. Office_Network

- **SSID Hidden Example**

- **Available Networks:**

1. Hidden Network
2. JioFiber_5G
3. Office_Network

- **Advantage of SSID Hiding**

- Casual users cannot easily see the Wi-Fi network in the list of available networks, providing a small layer of privacy.

- **Disadvantage of SSID Hiding**

- It does not provide real security because attackers can still detect the hidden SSID using Wi-Fi analysis tools.
- Users must manually enter the network name when connecting new devices.